MULTIMODAL AI

PROJECT REPORT

"Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis"

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted By

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1. Project Novelty

The common meeting summarisation tools available today use only the cloud-based generative language model. That decision presents two big issues in the corporate or legal world: the model itself can get its data wrong — it may "hallucinate" a decision or misattribute a quotation — and the voice itself must leave the premises and be stored in another company's servers, which likely isn't a desirable option for confidential boardroom discussions. The number of failures is minimized by avoiding the above two failure modes and hence, InfluenceIQ was created.

The other thing was, of course, training a custom model from scratch, and in fact ours was only about 100 hours worth in their AMI Meeting Corpus — that's not enough to train a competitive acoustic or language model. Instead we divided the problem in two parts and used the tools that were appropriate to each part: "train our own" and "hand everything to an LLM"

- Meetings: Perception (hearing the meeting): Faster-whisper (for transcription) and pyannote.audio (for diarisation), both in the inference mode. To put it into perspective: We have used only 100 hours of audio for pretraining Whisper, and already it gets transcription accuracy that our own 100 hours of audio would never have matched, let alone surpassed.
- Reasoning (understanding the meeting): a proprietary, deterministic, rule-based extraction engine instead of a generative model. Our pipeline reads the aligned discussion transcript and uses syntactic and keyword heuristics to extract the correct sentences where a proposal, objection, agreement, revision or action item took place, instead of asking an LLM to "summarise" the discussion.Because every line the system surfaces is a sentence that was literally spoken, at a literal timestamp, by a literal speaker, the output is 100% traceable back to the source audio — something a generative summary simply cannot promise. This is the core novelty we are pursuing: verifiable, non-hallucinating meeting intelligence, not another paraphrasing layer on top of an LLM.

Each line produced by the system is a true, literal sentence, spoken at a literal date and by a literal user, so there's a 100% traceability back to the source audio, which generative summaries can't offer. This is what we're working on in its essence, something that is not hallucinating but verifiable and intelligent/meaningful during the meeting, not another paraphrasing layer on top of an LLM.

The second novel idea is that 'the louder you speak, the more influence'. Everyone who attempts to determine who was responsible for the meeting defaults to the easiest to measure signal - airtime. We created an additional scoring system based on whether the contributions

of a person helped a decision, whether their objections were addressed, whether their revisions were adopted, whether their line helped to bring the line to a close, as opposed to whether they took the floor for a long or short period of time. This is more than a theoretical assertion—it is plainly reflected in the results of our own tests, as revealed in Section 3 below.

Lastly, the entire pipeline is intended to be operational on the outskirts. The whole perception layer is run on regular consumer devices (with the faster-whisper feature provided by CTranslate2) instead of the full-size models, and no audio will ever leave the machine on which it was recorded.

2. Methodology

The system is designed to be a four stage pipeline that converts a raw meeting recording (with or without a slide deck) into structured, evidence-based meeting intelligence. Each of the phases is intentionally designed to be independent, which means that if a component of it fails or has a problem, it does not silently cause another component to fail or have a problem in a later phase (for example, if the segment of the diarisation had a problem, then this does not affect the reasoning performed in a later phase).

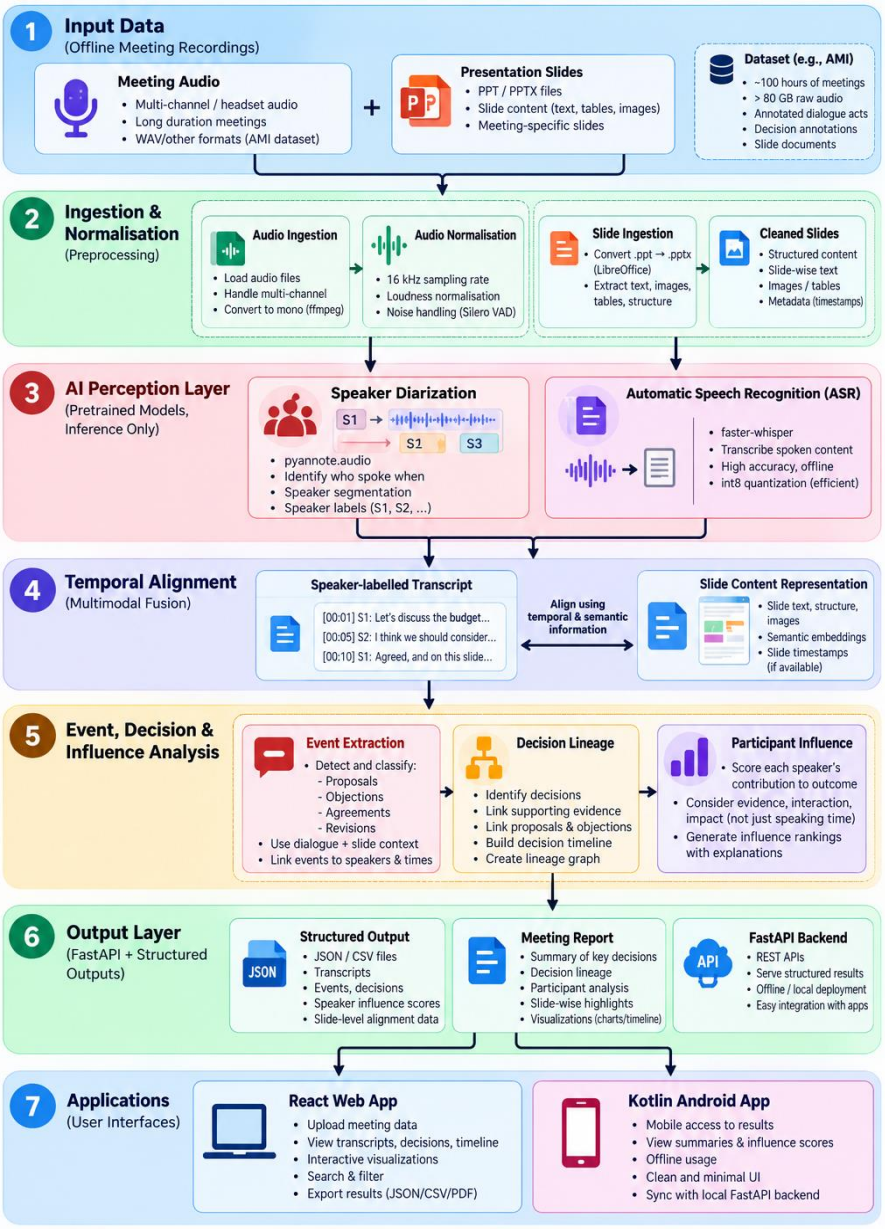


Figure 1: InfluenceIQ's four-phase offline pipeline, from raw audio to the React/Kotlin front ends.

2.1 Ingestion & Normalisation

Typical recordings are not clean — recordings on a dictaphone or phone on a boardroom table will have variable volume and whatever sample rate the device ends up using. All incoming data is processed by ffmpeg to be sure of a strict 16 kHz mono WAV format and a loudnorm filter to ensure that quieter speakers are played equally loudly to the downstream models as the loudest speaker in the room.

2.2 AI Perception Layer

The normalized sound is fed to two basic models simultaneously: pyannote.audio extracts the “who spoke when” information and faster-whisper translates the content. They are both used in inference mode only; that is, we load pre-trained weights and don't fine-tune them, hence we're able to achieve production grade accuracy with a small dataset. Once both outputs are available, we align text tokens mathematically to speaker boundaries, provided by Pyannote, to get a single speaker labelled transcript of Whisper.

2.3 Extractive NLP Engine (core contribution)

This is where the transcript gets converted into intelligence and is what we think is our intellectual contribution to the system. Three modules based on rules process the transcript sequentially, rather than passing it through an EM-based model:

- events.py — reads the transcript and tags a Proposal, Agreement, Objection and Revision as they are encountered using keyword and syntactic heuristics.
- decisions.py processes the weaving together of events, with regard to a topic; it keeps track of a topic from its first occurrence, through any objection/revision it gathers, until it reaches the point where it is confirmed as a final decision (if any).
- influence.py — this scores each player's interaction with that lineage, taking into account whether their revision was an accepted one or not, whether they said the final "accept" or not, etc., not just the number of words spoken.

2.4 Presentation Layer

The structured output – decisions, action items, participant scores and the evidence trail – is output as JSON and CSV, with a FastAPI backend to serve these outputs. The same meeting intelligence can be used in a React web app and a Kotlin Android app, which means that people can review results at their desks or right after the meeting on their phones.

3. First-Level Results

To validate the pipeline end-to-end, we ran it against meetings from the AMI corpus. The numbers below are directly from one representative test run (without any manual cleaning up), meeting_288ab746, and are what the system produced.

Metric	Value
Meeting analysed (test run)	AMI corpus — meeting_288ab746
Duration	00:35:33
Participants detected	6 (5 identified speakers + 1 unresolved segment)
Presentation slides linked	8
Raw conversational events detected	162
Confirmed decisions	1
Action items extracted	3
Tracked participant interactions	17

3.1 Decision Lineage Tracking

The system did not merely provide a summary of a topic's journey through the decision-making process to closure, but actually re-created the entire process: who initiated it, who opposed it, who edited it, or who finally agreed to close it. At this meeting, a single thread of discussion was followed through to a definite conclusion:

Decision ID	decision_001
Status	Confirmed
Topic / final text	“Either, whatever we decide the default to be…”
Participants involved	SPEAKER_04, SPEAKER_05, SPEAKER_00, SPEAKER_02, SPEAKER_01, UNKNOWN
Linked slide	slide_04

The full lineage the system reconstructed was: Then a question by SPEAKER_04 opened the topic, to which he raised an objection and SPEAKER_00 and SPEAKER_02 added objections, and there was finally a convergence on a common agreement, followed by a confirmed final decision by SPEAKER_00, who added the key revision. This is the type of chronological, evidence-based narrative that a summary would fail to keep a coherent thread on, such as here, where each step can be traced back to its speaker and the time of the speech.

3.2 Action Item Traceability

Being extractive, not generative, every action item that emerges from the pipeline is assigned to the speaker and the specific time and date that the action item was stated, nothing is assumed or paraphrased.

Action item (as spoken)	Speaker	Timestamp
"I'm just going to hopefully add something by next week that we can sort of try and…"	SPEAKER_00	00:20:16
"I'm not afraid to take that."	SPEAKER_02	00:31:59
"Like, take care of yourself."	SPEAKER_05	00:34:19

3.3 Participant Influence vs. Speaking Volume

How participants affect the speaker's voice.Participants' effects on the speaker's voice. This is the biggest finding from the first pass – this was the one we were testing our core hypothesis on – does the speaker that speaks the most drive the decision? The answer to the question is no, as indicated in the table and chart below.

Participant	Speaking Time	Speaking Rank	Influence Score	Influence Rank
SPEAKER_04	141.8 s	#1	0.3250	#2
SPEAKER_00	70.2 s	#2	0.3667	#1
SPEAKER_05	56.0 s	#3	0.0833	#5
SPEAKER_02	50.0 s	#4	0.1333	#3
SPEAKER_01	3.0 s	#6	0.1167	#4
UNKNOWN	6.0 s	#5	0.0167	#6

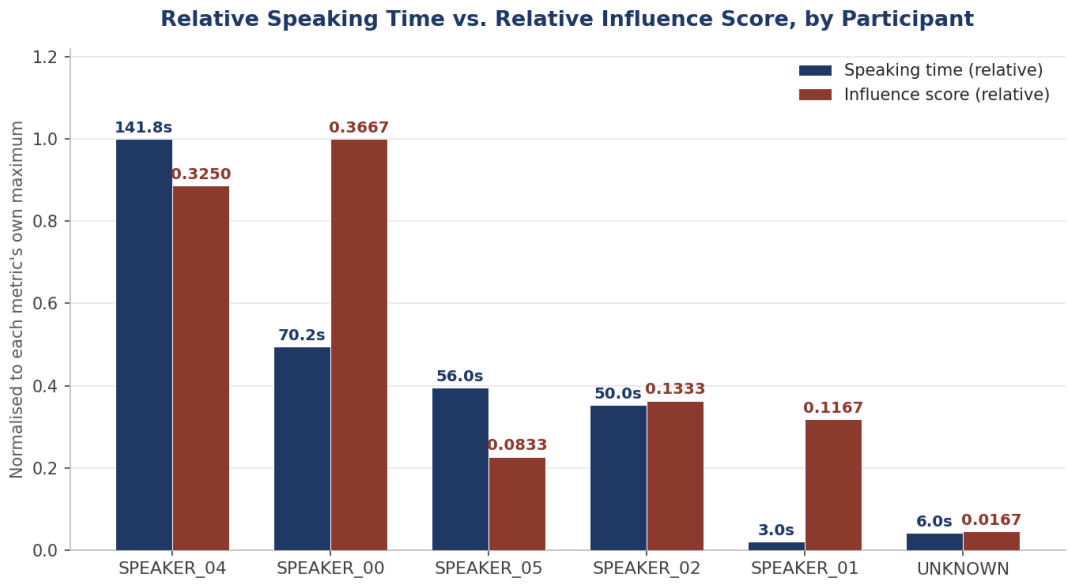


Figure 2: Relative speaking time vs. relative influence score, normalised to each metric's own maximum.

A pure acoustic analysis would have concluded that SPEAKER_04 held the meeting, as he led in both number of words spoken (141.8) and number of turns taken (66), more than double of any other speaker. We took the #2 spot in our influence engine. SPEAKER_00 speaks less (70.2 seconds), is credited with the adopted revision and the role of "final decision-maker" which makes them rank 1. The best example is SPEAKER_01, who was able to speak for just three seconds throughout two speaking turns, and yet outranked a speaker who spoke almost twice that long, SPEAKER_05, because SPEAKER_01's three seconds was directly anchored to a slide relevant to a decision.

We consider this as step one and not the end goal, as we need to see an adequate evaluation (a comparison of the system's rankings to human-annotated ground truth across multiple meetings) to be the next big step. Even at this point, this is a big enough disparity between the speaking-time ranking and the influence ranking to support the central thesis: They measure two different things, and a system that only measures the former is missing the point of the meeting.

4. Summary

It may seem like different varieties of the same theme, but the common thread throughout novelty, methodology, and this first round of results is this: create a system whose output can be verified against source recording, in other words, it's not something you have to take on faith. That's the key, and its a nice throw out to the two issues that enterprise and legal teams have had to contend with – hallucination and cloud data exposure – when considering the use of AI meeting tools – which are solved by using foundation models only for perception, while the reasoning remains strictly deterministic).